

MARBEFES

Stock Physical Accounts

Southern Gulf of Biscay BBT example

Basque Centre for Climate Change team

January 2025

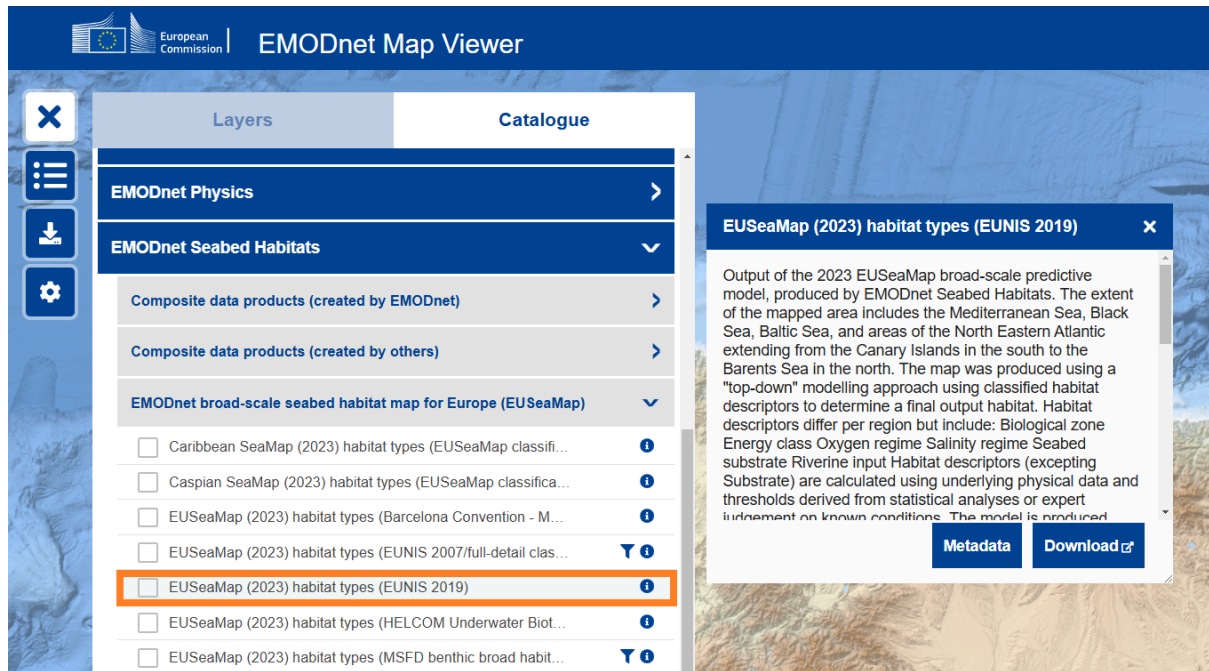
1. METHODS

Stock Physical Accounts in the Southern Gulf of Biscay BBT integrates the information of the Ecosystem Valuation Assessment (EVA) into level 3 EUNIS habitat classification (downloadable in [EMODnet Map Viewer](#)) to assess both ecosystem extent and condition. EUNIS habitat classification is used to calculate ecosystem extent while EVA is used to evaluate ecosystem condition. In the stock physical accounts, we use those EC (e.g. Benthic Habitats, Intertidal Macroalgae, Angiosperms) that can be directly linked to habitats, avoiding including those EC that can not be directly linked to habitats (e.g. Pelagic Fish, Seabirds). In this document, the steps that can be followed in QGIS and ArcGIS Pro software to compute Physical Accounts in the BBTs are described.

1.1. QGIS

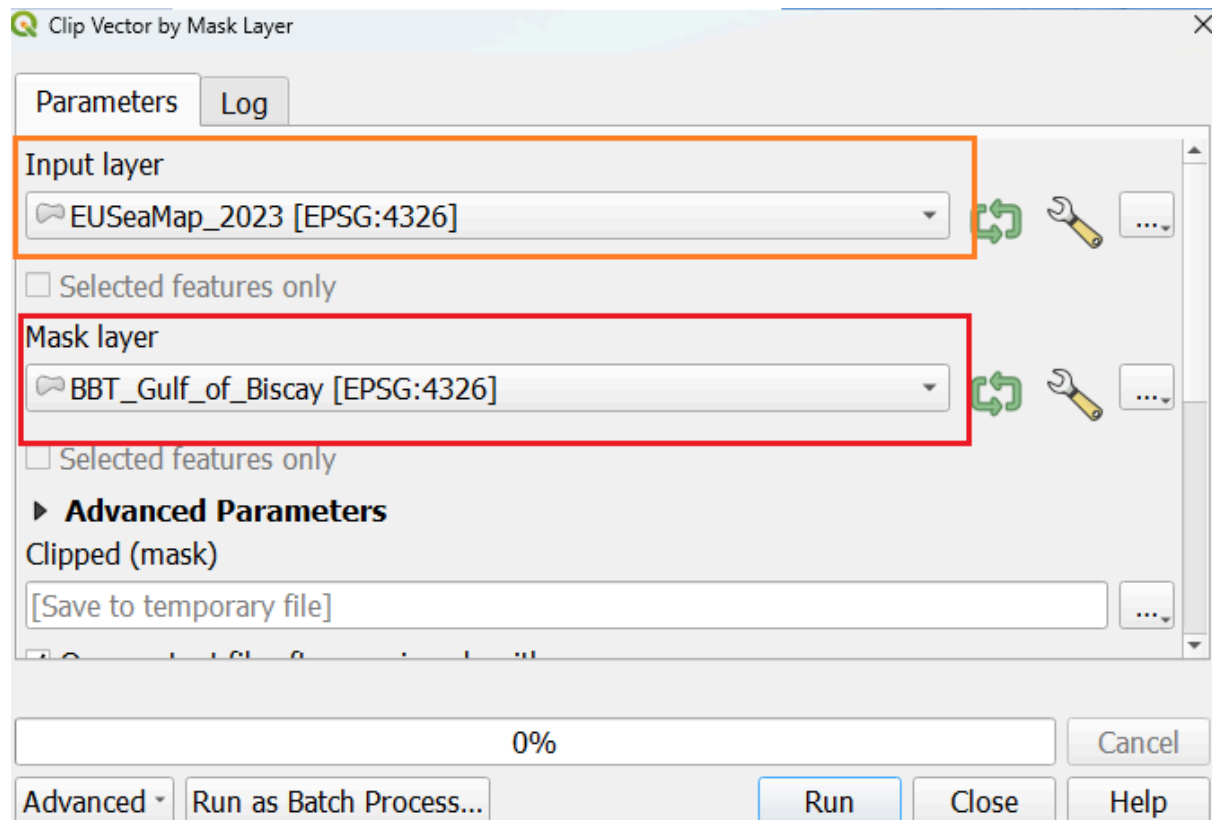
1.1.1. Download EUNIS habitat map

Download level 3 EUNIS habitat classification in [EMODnet Map Viewer](#) and unzip it in a folder. We can find it in the 'Catalogue' tab under the 'EMODNet Seabed Habitats' section. We select the third option 'EMODNet broad-scale seabed habitat map for Europe (EUSeaMap)' and we download the fifth option 'EUSeaMap (2023) habitat types (EUNIS 2019)'.



1.1.2. Clip EUNIS habitat to the BBT area

Clip the EUNIS habitat classification into the [Southern Gulf of Biscay BBT area](#). We can use the 'Clip vector by mask layer' tool to clip EUNIS to the BBT area. If you encounter any geometry problem with the EUNIS geometries, we can solve it using the 'Fix Geometries' tool. In the 'Clip vector by mask layer' tool we set the EUNIS habitat map as **Input layer** and the BBT shapefile as **Mask layer**.



1.1.3. Compute Ecological Values and Confidence of each EC

First of all, download the [Ecosystem Component \(EC\) EVAs](#). Then, create an average/max (depending on the old or new protocol) EV field using the 'Field Calculator' tool. We can access it right-clicking on the EVA vector file, we 'Open Attribute Table' and using 'Open Field Calculator' on the top-right side of the table. We can also use the SearchBar of the Processing Toolbox of QGIS (right panel). We define the new name (**Output field name**), type (**Output field type**), length (**Output field length**) and **expression** needed to compute the average/max EV.

BBT09_EVA_Angiosperms — Field Calculator

☐ Only update 0 selected feature(s)

☒ **Create a new field** ☐ **Update existing field**

☐ Create virtual field

Output field name: **EV_angios**

Output field type: 1.2 Decimal number (real)

Output field length: 10 Precision: 2

Expression: $("AQ3" + "AQ4" + "AQ5" + "AQ6" + "AQ7" + "AQ8" + "AQ9" + "AQ10" + "AQ11" + "AQ12" + "AQ13") / 11$

Function Editor: feature, geometry, id, NULL, 1.2 id, 1.2 AREA_m2, abc TYPE, 1.2 AQ3, 1.2 AQ4, 1.2 AQ5, 1.2 AQ6

Feature: COAST Preview: -9999

You are editing information on this layer but the layer is currently not in edit mode. If you click OK, edit mode will automatically be turned on.

OK Cancel Help

1.1.4. Clear the EVAs

Delete subzones (hexagons) with *NoData* values to avoid including them in the physical account analysis. We can use the 'Select features using an expression' tool to select *NoData*. Right-click on the EVA and open the attribute table. Find the 'Select features using an expression' tool in the top side of the table. Apply the **expression** to select *NoData* subzones, **enable 'Toggle Editing Mode'** and **Delete Selected Features**.

BBT09_EVA_Angiosperms — Select by Expression

Expression: **"EV_ANGIOSP" = -9999**

NoData code

Function Editor: feature, geometry, id, Aggregates, Arrays, Color, Conditionals, Conversions, Custom, Date and Time, Fields and Values, File and Paths

Feature: COAST Preview: 1

Help Select Features Close

BBT09_EVA_Angiosperms — Features Total: 5285, Filtered: 5285, Selected: 4354

1.2 id = € 1.2 Update All Update Selected

	id	AREA_m2	TYPE	AQ3	AQ4	AQ5
1	39360	54126.58773650...	ESTUARY	5.000000000000	1.95431042515	5.000000000000
2	14674	54126.58773650...	ESTUARY	0	0	0
3	14673	54126.58773650...	ESTUARY	0	0	0
4	14662	54126.58773650...	ESTUARY	0	0	0
5	14663	54126.58773650...	ESTUARY	0	0	0
6	14664	54126.58773650...	ESTUARY	0	0	0
7	14665	54126.58773650...	ESTUARY	0	0	0

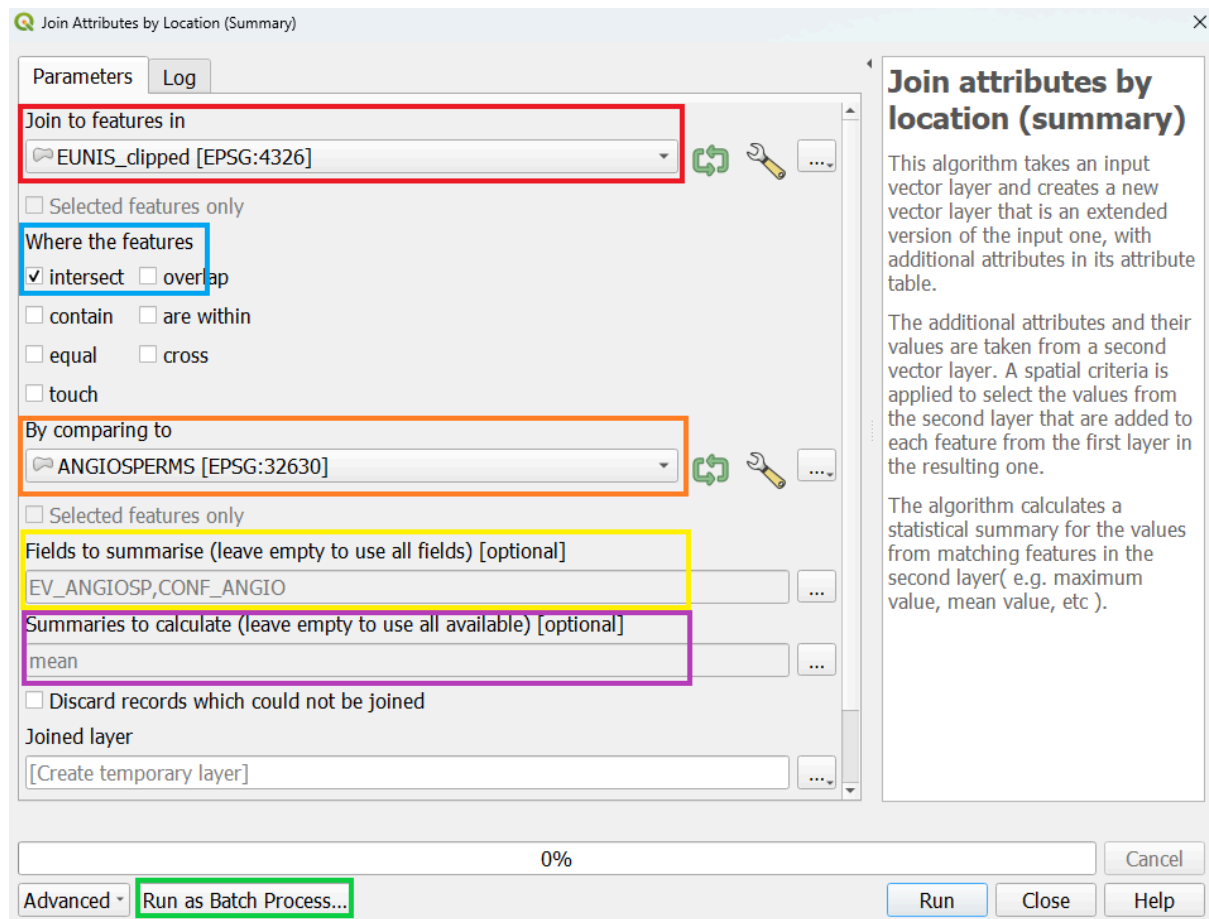
Show All Features

1.1.5. Compute EV in all ECs and clean them

Repeat the steps 1.1.3 and 1.1.4 to compute EV for all ECs and clean *NoData* subzones.

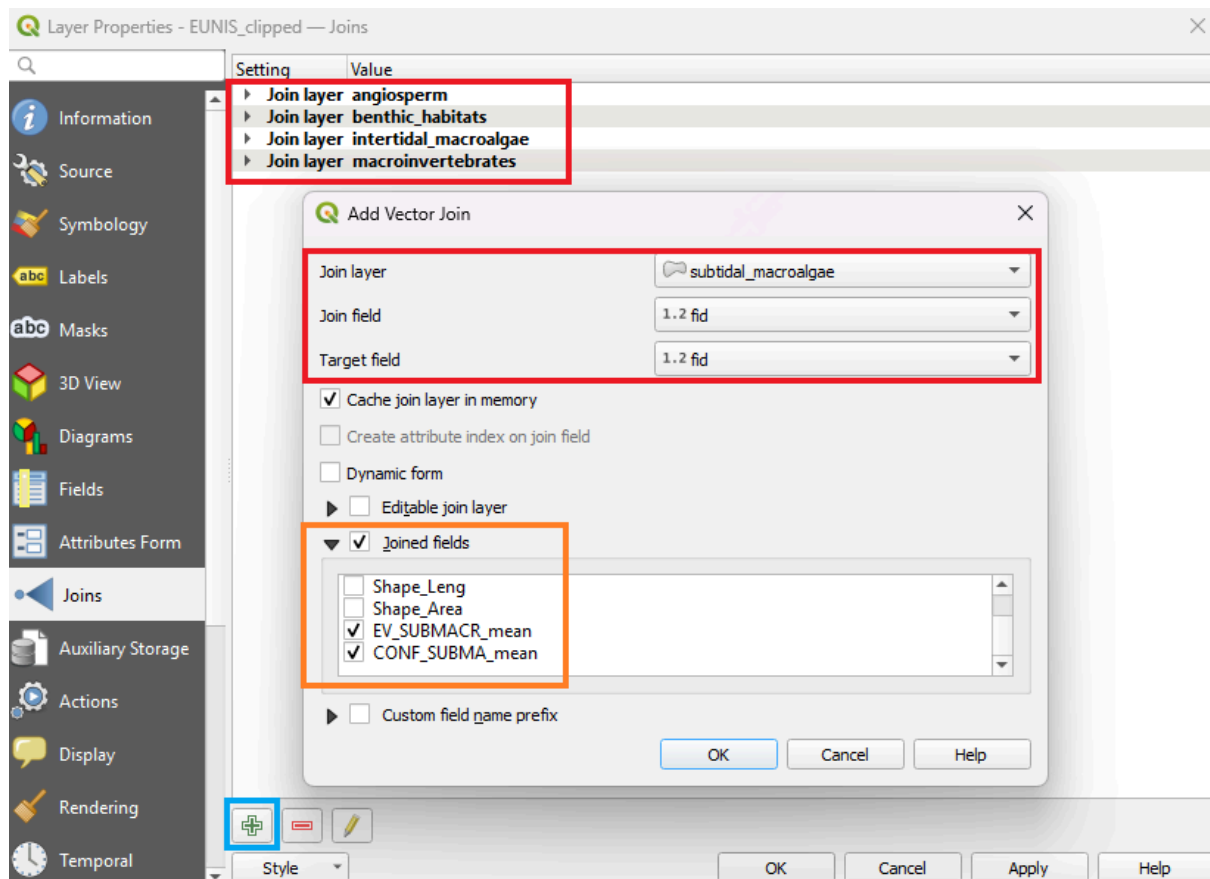
1.1.6. Integrate EVAs into EUNIS habitat maps

Compute the EV and Confidence of each EUNIS habitat polygon using the **average EV and Confidence scores** of ECs. Each hexagon of the **ECs** that **intersect** with each of the **EUNIS habitat polygons** will be taken into account. We can use the 'JoinAttributes by location (summary)' tool to do so. Repeat the process with all EC EVAs to add mean EV and Confidence scores into the EUNIS habitat file or '**Run as Batch Process**' to do so.

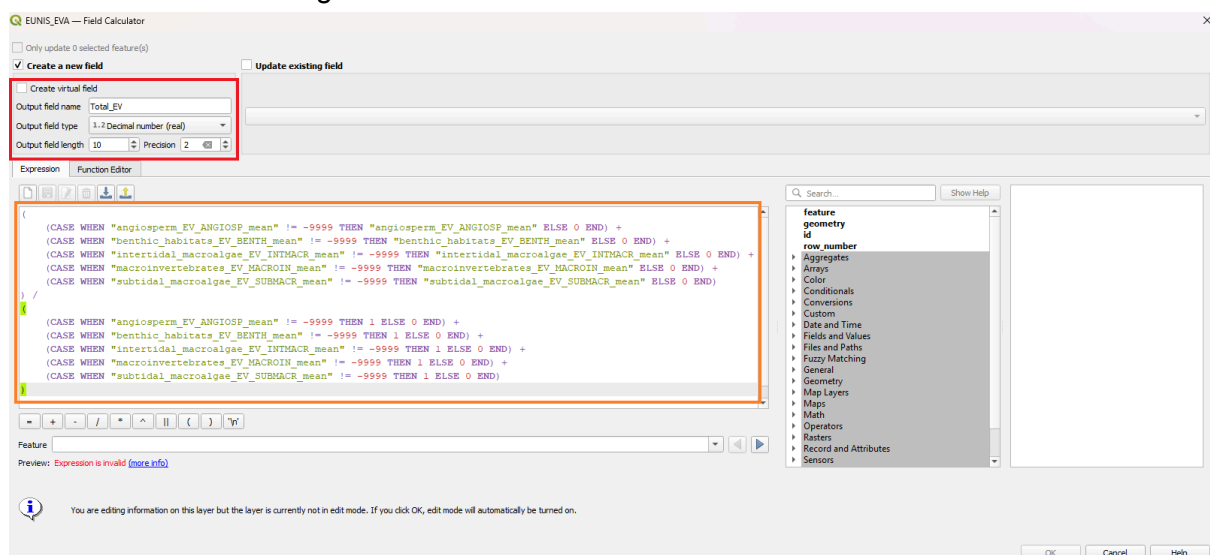


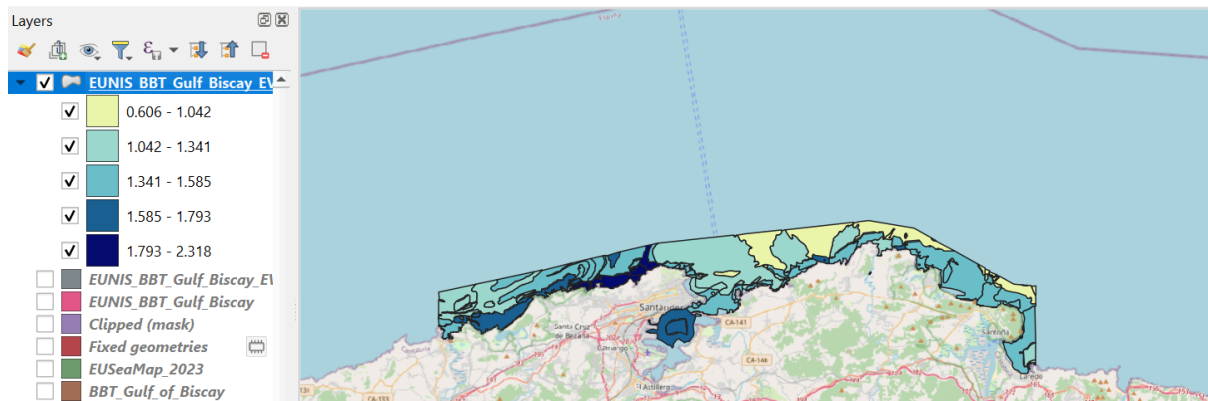
1.1.7. Compute Overall habitat EV and Confidence scores

First of all, we need to join the EV and Confidence scores obtained in step 1.1.6 into a single file. To do so we can do a Join between all – EUNIS habitat-EC EV and Confidence – files. Right-click on the EUNIS clipped file, select 'Properties' and go to 'Joins'. 'Add new joins' to incorporate the EV and Confidence scores of each – EUNIS habitat-EC EV and Confidence – files into the EUNIS clipped file.



Then, we compute overall EV and Confidence scores for each EUNIS habitat polygon. We use the 'Field Calculator' to obtain the **Total_EV** and **Total_CONFIDENCE** of each EUNIS habitat polygon using the following **expression** with the corresponding field names and **NoData** codes. As a result, we obtain the EUNIS habitat map with a field (**Total_EV**) ranking the the areas with the highest EV.





1.1.8. Group EV and Confidence by EUNIS habitat type

Compute area-weighted EV and Confidence scores based on EUNIS habitat types (“EUNIS2019D”) and areas (“area”) of each polygon. First, open the ‘Field Calculator’ to create a new field called ‘area’ and use the expression **\$area** to compute the area of each EUNIS habitat polygon (notice that if the coordinate system of the EUNIS habitat file is geographic, you must reproject the file into a projected coordinate system or compute the geodesic area). Then, use the following **expression** with the corresponding field names to compute **area-weighted EV and Confidence scores for each EUNIS habitat type**.

Only update 0 selected feature(s)

☒ Create a new field

☐ Update existing field

☐ Create virtual field

Output field name: Habitat_EV

Output field type: 1.2 Decimal number (real)

Output field length: 10 Precision: 2

Expression: `sum("Total_EV"* "area", group_by:= "AllcombD") / sum("area", group_by:= "AllcombD")`

Feature:

Preview: Expression is invalid ([more info](#))

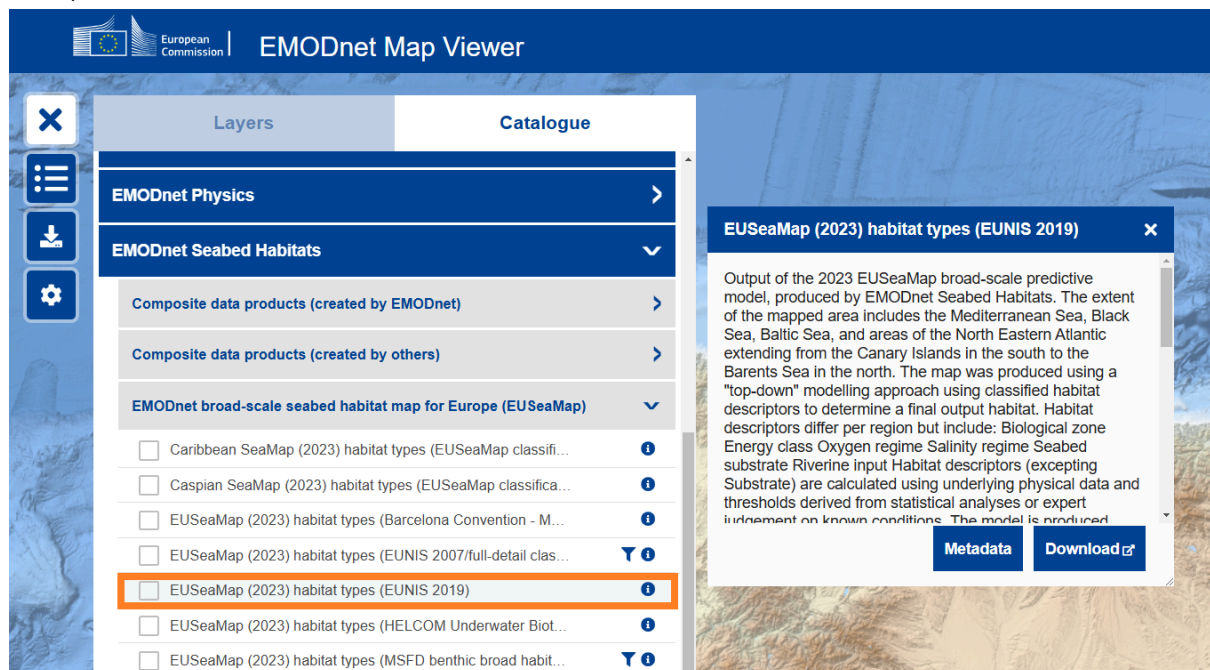
You are editing information on this layer but the layer is currently not in edit mode. If you click OK, edit mode will automatically be turned on.

OK Cancel Help

1.2. ArcGIS Pro

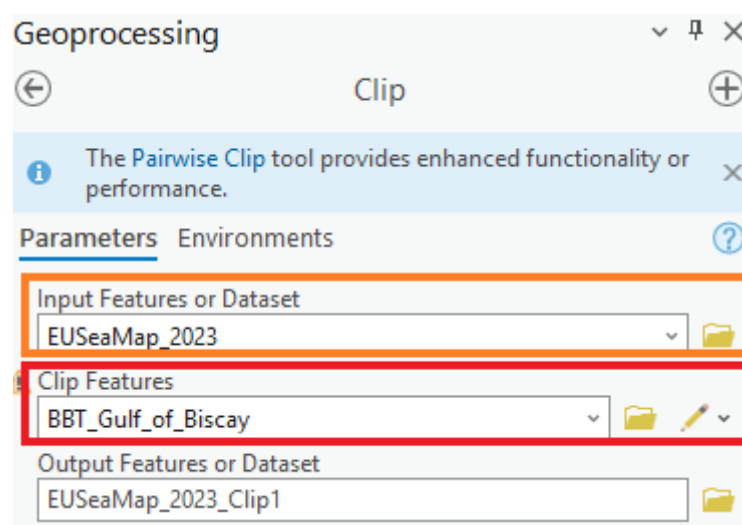
1.2.1. Download EUNIS habitat map

Download level 3 EUNIS habitat classification in [EMODnet Map Viewer](#) and unzip it in a folder. We can find it in the 'Catalogue' tab under the 'EMODNet Seabed Habitats' section. We select the third option 'EMODNet broad-scale seabed habitat map for Europe (EUSeaMap)' and we download the fifth option 'EUSeaMap (2023) habitat types (EUNIS 2019)'.



1.2.2. Clip EUNIS habitat to the BBT area

Clip the EUNIS habitat classification into the [Southern Gulf of Biscay BBT area](#). We can use the 'Clip' tool from the Geoprocessing panel to clip EUNIS to the BBT area. In the 'Clip' tool we set the EUNIS habitat map as **Input Features or Dataset** and the BBT shapefile as **Clip Features**.



1.2.3. Compute Ecological Values and Confidence of each EC

First of all, download the [Ecosystem Component \(EC\) EVAs](#). Then, create an average/max (depending on the old or new protocol) EV field using the 'Calculate field' tool. First, we create new fields using the 'Add Field' on the top-left side of the table. Then, we can access the 'Calculate Field' tool by right-clicking on the EVA vector file, 'Attribute Table' and using 'Calculate Field' (select the created new field and right-click). We can also use the SearchBar of the Geoprocessing panel of ArcGIS Pro (right panel). We define the **expression** needed to compute the average/max EV.

Attribute Table: BBT09_EVA_Benthic_Habitat

FID	Shape	AQ1
1	65	5
2	252	5
3	334	5
4	340	5
5	406	5
6	423	5
7	505	5
8	745	5
9	757	5
10	759	5
11	859	5
12	919	5

Fields View: BBT09_EVA_Benthic_Habitat (BBT S)

Visible	Read Only	Field Name	Alias	Data Type
☒	☐	AQ6	AQ6	Double
☒	☐	AQ7	AQ7	Double
☒	☐	AQ8	AQ8	Double
☒	☐	AQ9	AQ9	Double
☒	☐	AQ10	AQ10	Double
☒	☐	AQ11	AQ11	Double
☒	☐	AQ12	AQ12	Double
☒	☐	AQ13	AQ13	Double
☒	☐	TOTAL_EV	TOTAL_EV	Double
☒	☐	FIELD_DATA	FIELD_DATA	Long
☒	☐	CONFIDENCE	CONFIDENCE BENTHIC HABITAT	Double
☒	☐	Avg_EV	AVERAGE EV BENTHIC HABITAT	Double

Click here to add a new field.

Avergae EV Intertidal Macroalgae	
3	
3	
3	
7	
7	
7	
7	
7	
7	
7	
4	
4	
4	
4	
4	
4	
4	2.031
4	2.892
4	2.65
4	2.65
4	2.65
4	2.031
4	2.892

Sort Ascending
 Sort Descending
 Custom Sort Ctrl+Shift+S

Hide Field
 Freeze/Unfreeze Field
 Calculate F*ie*ld

Calculate Get Values
 Statistics
 Summarize
 Fields
 Delete

Calculate Field
 Set the values of this field by specifying a calculation expression. If any of the rows in the table are currently selected, only the values of the selected rows will be calculated.

Calculate Field

This tool modifies the Input Table

Input Table

BBT09_EVA_IntMacroalgae

Field Name (Existing or New)

Avergae EV IntertidalMacroalgae

Expression Type

Python 3

Expression

Fields

AQ9

AQ10

AQ11

AQ12

AQ13

TOTAL_EV

FIELD_DATA

CONFIDENCE

Helpers

.as_integer_ratio()

.capitalize()

.center()

.conjugate()

.count()

.decode()

.denominator()

Insert Values

AvgEV_IMac =

(!AQ7!+!AQ8!+!AQ9!+!AQ10!+!AQ11!+!AQ12!+!AQ13!)/7

Code Block

Enable Undo

Apply

OK

1.1.4. Clear the EVAs

Delete subzones (hexagons) with *NoData* values to avoid including them in the physical account analysis. We can use the 'Select By Attributes' tool to select *NoData*. Right-click on the EVA and open the attribute table. Find the 'Select By Attributes' tool on the top side of the table. Apply the **expression** to select *NoData* subzones and 'Delete Selected Features'.

Select By Attributes ? x

Input Rows
BBT09_EVA_IntMacroalgae

Selection Type
New selection

Expression
Load Save Remove

SQL ☐

Where TOTAL_EV is equal to -9999

+ Add Clause

☐ Invert Where Clause

Apply OK

BBT09_EVA_Benthic_Habitat x

Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy

	FID	Shape *	id	AREA_m2	TYPE	AQ1	AQ2	AQ3	AQ4	AQ5	AQ6	AQ7	Delete Selection (Delete)			
1	65	Polygon	63770	54126.587736	COAST	5	0.928753	5	0.928753	5	0.928753	0.909091	Delete selected rows from the table.			
2	252	Polygon	62990	54126.587736	COAST	5	1.303012	5	1.303012	5	1.303012	0.909091	0.065645	0.065645	0	0
3	334	Polygon	71596	54126.587736	COAST	5	2.803214	5	2.803214	5	2.803214	0.909091	0.09224	0.09224	0	0
4	340	Polygon	71595	54126.587736	COAST	5	0.423965	5	0.423965	5	0.423965	0.909091	0.378552	0.378552	0	0
5	406	Polygon	63249	54126.587736	COAST	5	0.676601	5	0.676601	5	0.676601	1.363636	0.228512	0.228512	1.25	0.032518
6	423	Polygon	21666	54126.587736	COAST	5	1.777044	5	1.777044	5	1.777044	1.363636	0.311631	0.311631	1.25	0.006259
7	505	Polygon	70803	54126.587736	COAST	5	0.243601	5	0.243601	5	0.243601	0.909091	0.04717	0.04717	0	0
8	745	Polygon	62729	54126.587736	COAST	5	2.357177	5	2.357177	5	2.357177	0.909091	0.258061	0.258061	0	0
9	757	Polygon	21148	54126.587736	COAST	5	4.880499	5	4.880499	5	4.880499	0.909091	0.097645	0.097645	0	0
10	759	Polygon	21147	54126.587736	COAST	5	0.04989	5	0.04989	5	0.04989	1.363636	0.488624	0.488624	1.25	0.000313
11	859	Polygon	70283	54126.587736	COAST	5	0.962048	5	0.962048	5	0.962048	0.909091	0.023896	0.023896	0	0
12	919	Polygon	21407	54126.587736	COAST	5	1.579557	5	1.579557	5	1.579557	0.909091	0.228408	0.228408	0	0

1.1.5. Compute EV in all ECs and clean them

Repeat the steps 1.2.3 and 1.2.4 to compute EV for all ECs and clean *NoData* subzones.

1.1.6. Integrate EVAs into EUNIS habitat maps

First, we create in the EUNIS habitat file new *double* fields where average EV & Confidence scores will be stored. Create one EV & Confidence field for each Ecosystem Component.

Fields: EUNIS_BBT_So...Gulf of Biscay) X

Current Layer: EUNIS_BBT_Southern_Gulf_Biscay (▾)

Visible	Read Only	Field Name	Alias	Data Type	Allow NULL
☒	☐	EV_Angio	EV Angiosperm	Float	☐
☒	☐	CONF_Angio	CONFIDENCE Angiosperm	Float	☐
☒	☐	EV_Benth	EV Benthic Habitats	Float	☐
☒	☐	CONF_Benth	CONFIDENCE Benthic Habitats	Float	☐
☒	☐	EV_MInv	EV MacroInvertebrates	Float	☐
☒	☐	CONF_MInv	CONFIDENCE MacroInvertebrates	Float	☐
☒	☐	EV_IMacr	EV IntertidalMacroalgae	Float	☐
☒	☐	CONF_IMacr	CONFIDENCE IntertidalMacroalgae	Float	☐
☒	☐	EV_SMacr	EV SubtidalMacroalgae	Float	☐
☒	☐	CONF_SMacr	CONFIDENCE SubtidalMacroalgae	Float	☐

Then, we compute the EV and Confidence of ECs using the average EV and Confidence of all ECs. Each hexagon of the ECs that fall into each of the EUNIS habitat polygons will be taken into account. Do a 'Spatial Join' between the EUNIS habitat file and each EVA shapefiles. Define **Match Option** as **Intersect** and **Merge Rule** of EV and Confidence fields as **Mean**.

The screenshot shows the 'Add Spatial Join' dialog box in ArcGIS Pro. The 'Target Features' is 'EUNIS_Gulf_Biscay_BTT' and the 'Join Features' is 'BBT09_EVA_Angiosperms'. The 'Match Option' is set to 'Intersect'. The 'Fields' section shows 'CONFIDENCE' and 'Avg_EV' with a 'Merge Rule' of 'Mean'. The 'Search Radius' is set to 'Decimal Degrees'. The 'Output Fields' section shows 'CONFIDENCE' and 'Avg_EV' with a 'Merge Rule' of 'Mean'.

Copy average EV and Confidence results into the fields created in the beginning of this step (If there is no hexagons intersecting an EUNIS habitat polygon ArcGIS Pro defines the value as null and then, when we copy the result into the fields created in *Step g*, this null values are set to 0).

Repeat the steps described in 1.2.6 with all EC EVAs to add average EV and Confidence information to the EUNIS habitat file.

1.1.7. Compute Overall habitat EV and Confidence scores

Compute the overall EV & Confidence scores of each EUNIS polygon using the 'Field Calculator'. Use the following expressions with the corresponding field names to avoid including *NoData* values in the computation. First, create two *double* new fields to store the TOTAL_EV & TOTAL_CONFIDENCE.

Calculate Field

This tool modifies the Input Table

Input Table: EUNIS_BBT_Southern_Gulf_Biscay

Field Name (Existing or New): Overall EV

Expression Type: Python 3

Expression

Fields

- CONFIDENCE Benthic Habitats
- EV MacroInvertebrates
- CONFIDENCE MacroInvertebrates
- EV IntertidalMacroalgae
- CONFIDENCE IntertidalMacroalgae
- EV SubtidalMacroalgae
- CONFIDENCE SubtidalMacroalgae
- Overall EV

Helpers

- .as_integer_ratio()
- .capitalize()
- .center()
- .conjugate()
- .count()
- .decode()
- .denominator()

Insert Values

Overall_EV =

compute_mean(!EV_Angio!, !EV_Benth!, !EV_MInv!, !EV_IMacr!, !EV_SMacr!)

Code Block

```
def compute_mean(*values, nodata_value = 0):
    # Filter valid values
    valid_values = [v for v in values if v != nodata_value]
    # Computes mean if there are valid values
    if valid_values:
        return sum(valid_values)/ len(valid_values)
    # If there are not valid values return 0
    return 0
```

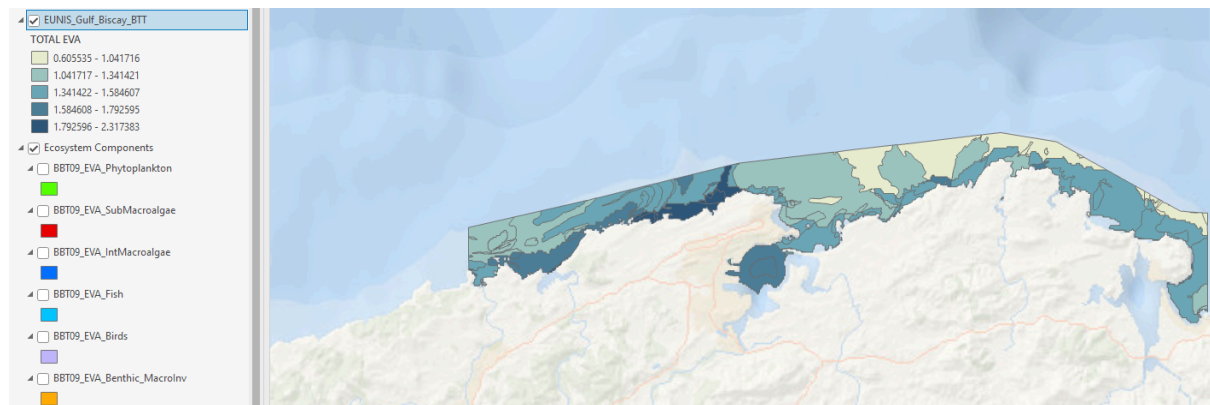
✓ Expression is valid

☐ Enforce Domains

Enable Undo ☐

Apply OK

As a result, we obtain the EUNIS habitat map with a field (TOTAL_EV) ranking the areas with the highest EV.



1.1.8. Group EV and Confidence by EUNIS habitat type

Compute area-weighted EV and Confidence scores based on EUNIS habitat types (“**EUNIS2019D**”) and areas (“area”) of each polygon. First, create a new field to store the area of each polygon and use ‘Calculate Geometry’ to compute the area of each EUNIS habitat polygon (use ‘area(geodesic)’ if the coordinate system is geographic (e.g. WGS84)). Then, create two new fields to store Grouped_EV (equals to Habitat_EV in the QGIS guide) & Grouped_CONFIDENCE (equals to Habitat_CONF in the QGIS Guide). Compute them using ‘Calculate Field’ tool and the following **expression**.

Calculate Field

This tool modifies the Input Table

Input Table
EUNIS_BBT_Southern_Gulf_Biscay

Field Name (Existing or New)
Grouped EV

Expression Type
Python 3

Expression

Fields

FID
Shape
ModelCode
Oxygen
Salinity
Energy
Biozone

Helpers

as_integer_ratio()
capitalize()
center()
conjugate()
count()
decode()
denominator()

Insert Values

GroupEV =
compute_weighted(!AllcombD!)

Code Block

```
def compute_weighted(habitat):
    import arcpy
    where_clause = f"!AllcombD = '{(habitat)}'"
    with arcpy.da.SearchCursor(f"C:\\Users\\beñat.egidazu\\Desktop\\PhD\\Papers\\Physical_Accounts\\Souther_Gulf_Biscay\\EUNIS\\EUNIS_BBT_Southern_Gulf_Biscay.shp", ["Overall_EV", "area"], where_clause) as cursor:
        numerator = 0
        denominator = 0
        for row in cursor:
            numerator += row[0] * row[1] # EV * area
            denominator += row[1] # Sum of area
        if denominator == 0:
            return 0
        return numerator / denominator
```

Enforce Domains

Enable Undo ☐ Apply OK